AWS VPC PEERING PROJECT

Private Connectivity Between Two Amazon VPCs

Submitted by: Anmol Kumar Singh

Course: EWB Courses — AWS Networking

AWS Region: Asia Pacific (Mumbai) — `ap-south-1`

1. Introduction

This project demonstrates private communication between two Amazon Virtual Private Clouds (VPCs) using an AWS VPC peering connection. It explores routing, security groups, private EC2 instances, and secure access through a bastion host.

2. Objectives

- Create two VPCs with different IP address ranges.
- Configure public and private subnets.
- Establish a VPC peering connection.
- Configure route tables and security groups.
- Test communication between private EC2 instances.

3. Network Configuration

Component	CIDR / Private IP
VPC A	10.10.0.0/16
VPC A Public Subnet	10.10.1.0/24
VPC A Private Subnet	10.10.2.0/24
VPC B	10.20.0.0/16
VPC B Private Subnet	10.20.1.0/24
App-A	10.10.2.192
App-B	10.20.1.96

4. Implementation

A VPC peering connection was configured between VPC A and VPC B. Route tables were configured to direct traffic between the two networks through the peering connection. Security groups controlled SSH and ICMP traffic. A bastion host provided secure SSH access to the private instances.

5. Testing

The lab included the following connectivity tests:

- Ping from App-A to App-B.
- Ping from App-B to App-A.
- SSH access to App-B through the bastion host.
- Controlled failure and recovery by changing an ICMP security-group rule.

The recorded lab session reported successful bidirectional ping tests with zero packet loss and successful SSH access through the bastion host.

6. Security and Cleanup

The application instances used private IP addresses without requiring public IPv4 addresses. Security groups restricted access to the required traffic. After completing the lab, the EC2 instances, VPCs, peering connection, and associated lab resources were removed or checked during cleanup.

7. Conclusion

This project provided practical knowledge of AWS VPC peering, route tables, security groups, private EC2 instances, and bastion-host access. It demonstrated that private communication requires correct routing in both directions and appropriate security-group rules.